

Real-Time Outlier Detection with Dynamic Process Limits

1st Marek Wadinger

*Institute of Information Engineering, Automation and
Mathematics
Slovak University of Technology in Bratislava
Bratislava, Slovakia
marek.wadinger@stuba.sk*

2nd Michal Kvasnica

*Institute of Information Engineering, Automation and
Mathematics
Slovak University of Technology in Bratislava
Bratislava, Slovakia
michal.kvasnica@stuba.sk*

Abstract—Anomaly detection methods are part of the systems where rare events may endanger an operation’s profitability, safety, and environmental aspects. Although many state-of-the-art anomaly detection methods were developed to date, their deployment is limited to the operation conditions present during the model training. Online anomaly detection brings the capability to adapt to data drifts and change points that may not be represented during model development resulting in prolonged service life. This paper proposes an online anomaly detection algorithm for existing real-time infrastructures where low-latency detection is required and novel patterns in data occur unpredictably. The online inverse cumulative distribution-based approach is introduced to eliminate common problems of offline anomaly detectors, meanwhile providing dynamic process limits to normal operation. The benefit of the proposed method is the ease of use, fast computation, and deployability as shown in two case studies of real microgrid operation data.

Index Terms—anomaly detection, interpretable machine learning, online machine learning, real-time systems, streaming analytics

I. INTRODUCTION

The era of Industry 4.0 is ruled by data. Effective data-based decision-making is driven by the quantity of collected data. Internet of Things (IoT) devices made data acquisition seamless and positively influenced a wide range of industries. It is estimated that the annual economic impact of IoT will further grow and reach up to \$6.2 trillion by 2025 [1].

Various data collection mechanisms are used to buffer and store the data for future processing. However, the tremendous increase in data availability and the desire to extract valuable insight led to problems with unbounded buffering.

Streaming data analytics introduced mechanisms for online extraction and transformation while loading to the storage only a fraction of the former data load, which allowed the storage of the vital information carried by the data more comprehensively. However, the unstable quality of the data appeared to have the most crucial importance over the quantity.

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Anomaly detection, well studied in the last decades, was reborn to the world of new challenges. Former studies were mainly concerned with a domain-specific detection of various anomalies while trained offline [2]. However, anomalies of diverse sources, from fraudulent web activity to sensor failure, malfunctioning of the hardware, and performance drops, mutate over time, and the model had to be updated.

Companies expanded their research activities on the creation and integration of generic frameworks combining prediction, detection, and alert mechanisms. One of the first projects, open-sourced for the public, are EGADS by Yahoo [3] and AnomalyDetection by Twitter [4]. The frameworks’ modularity allowed the automation of the anomaly detection of time-series data and created space for discussion.

Moving from domain-specific to generic methods posed new problems connected to type I errors, i.e., a false-positive classification of normal behavior as anomalous. Accurate selection of forecaster, detector, and alerting mechanism allowed to tackle the problem, nevertheless, introduced considerable dependence on expert domain knowledge and fine-tuning.

Further work proved improvement in performance while relieving the tight requirements on domain knowledge [5]. However, strict demands on detection systems ranging from lasting up times to continuous monitoring with stable performance pointed to the challenge of data stationarity. Change points and concept drift troubled unsupervised models, which led to service downtime due to the model retraining.

The era of adaptive machine learning introduced incremental learning schemes as a solution. Multiple studies for learning modes, adaptation methods, and model management swept through the machine learning community. Pannu et al. proposed an adaptive anomaly detection system [6]. However, the method represented a supervised operator-in-the-loop solution. Zhang et al. introduced an adaptive kernel density-based algorithm that uses an adaptive kernel width [7]. Nonetheless, training the models on big data had limitations resulting from storing and unbounded data buffering. Online learning models relaxed the need for data availability during model training [8]. On the contrary, it processed the data from a bounded buffer sequentially as in [9] and [10].

Anomaly detection in microgrids, however, called for low

latency detection which implied real-time training and prediction processes [11]. Such adaptation of streamed modeling took into consideration strict boundaries on computational time. For work in this area see [12] and [13].

Alerting mechanisms in process automation detect situations where signal value deviates from constraints. An alert watchdog is triggered on threshold violation by individual signals. The constraints, or process limits, are usually predefined and fixed. Nevertheless, factors such as aging and environmental changes call for dynamic process limits. Setting up a procedure for an evergrowing number of signal measurements is time-consuming. Besides, it is impossible for signals where no prior information about a correct process range is known. Those are subject to external factors that are unknown at setup time.

In this article, we suggest using existing process automation infrastructure based on alerting (PLC, SCADA, among others) and applying machine learning for dynamic process range based on changing conditions. We propose an unsupervised anomaly detection algorithm capable of online adaptation to change points and concept drifts, which adds to a recently developed body of research. The approach is evaluated on two case studies of microgrid sensors. To the author's knowledge, there are no studies to date concerned with providing adaptive operation constraints.

The main benefits of the proposed solution are that it:

- Keeps existing IT infrastructure, saving costs, and does not require operator retraining
- Automates alerting thresholds setup for a high number of signals
- Automates alerting for signals with no a priori knowledge of process limits
- Assesses changing environmental conditions and device aging
- Uses self-learning approach on streamed data

II. PRELIMINARIES

This section introduces the main concepts which are building pillars of the developed approach. Subsection II-A will discuss a one-pass algorithm that allows for online adaptation. The following Subsection II-B proposes the ability to invert the solution in a two-pass implementation. The mathematical background of distribution modeling in Subsection II-C provides a basis for the Gaussian anomaly detection model conceptualized in the last Subsection II-D of Preliminaries.

A. Welford's Method

Streaming data analytics restricts the uncontrolled growth of memory usage by keeping only the data required for computations. One-pass algorithms allow processing on-the-fly without storing the entire data stream.

Definition 2.1 (One-pass algorithm): The algorithm with a single access to the data items in the order of their occurrence, i.e., $x_1, x_2, x_3, \dots$ is called one-pass algorithm [14]

Welford's method represents a numerically stable one-pass solution for the online computation of mean and variance [15].

Given x_i where $i = 1, \dots, n$ is the sample index in given population n , the corrected sum of squares is defined as

$$S_n = \sum_{i=1}^n (x_i - \bar{x}_n)^2, \quad (1)$$

where the running mean $\bar{x}_n$ is

$$\bar{x}_n = \frac{n-1}{n} \bar{x}_{n-1} + \frac{1}{n} x_n = \bar{x}_{n-1} + \frac{x_n - \bar{x}_{n-1}}{n}. \quad (2)$$

The following identities to update the corrected sum of squares hold true

$$S_n = S_{n-1} + (x_n - \bar{x}_{n-1})(x_n - \bar{x}_n), \quad (3)$$

and the corresponding variance is

$$s_n^2 = \frac{S_n}{n-1}. \quad (4)$$

As we can see in (3), we do access only current data sample x_n and previous value of $\bar{x}_{n-1}$ which is updated in (2) using the same data sample and the size of seen population n .

B. Inverse Welford's Method

Let the incoming stream of data be subject to the concept drift. Such alternation in statistical properties has a negative influence on prediction accuracy. An adaptation of any machine learning model is crucial for successful long-term operation.

Definition 2.2 (Concept drift): Concept drift is a change in the statistical properties that occur in a sub-region of the feature space.

The previous Subsection II-A defined the main concept of online statistical computation that allows reacting to such changes. However, the further in time the shift occurs, the slower the adjustment of the running mean is, resulting from a negative relationship in (2) between population size n and influence of the last sample in population x_n on the updated value of $\bar{x}_n$. For this reason, we define the expiration period t_e , over which the running statistics are computed. After the expiration period, the data items are forgotten. Such reversal results in a need to store all the data in the window in order to revert their effect. Given $t_e = n-1$ we can revert the influence of the first data sample on the running mean as

$$\bar{x}_{n-1} = \frac{n}{n-1} \bar{x}_n - \frac{1}{n-1} x_{n-t_e} = \bar{x}_n - \frac{x_{n-t_e} - \bar{x}_n}{n-1}, \quad (5)$$

then reverting the sum of squares follows as

$$S_{n-1} = S_n - (x_{n-t_e} - \bar{x}_{n-1})(x_{n-t_e} - \bar{x}_n), \quad (6)$$

which allows the computation of variance

$$s_{n-1}^2 = \frac{S_{n-1}}{n-2}. \quad (7)$$

C. Modeling Distribution

Statistical distribution can be used to create a generalized model of a normal system behavior based on observed measurement. Specifically, in cases where a change point is not anticipated within a given subset of samples, we make the assumption that the data conforms to a Gaussian normal distribution. Parameters of the normal distribution are used to compute standard score for each new observation. Standard score z_i specifies the number of sample standard deviations s_n^2 by which x_i deviates from mean $\bar{x}_n$ of normal distribution

$$z_i = \frac{x_i - \bar{x}_n}{s_n^2}. \quad (8)$$

To compute the general probability of z_i belonging to anomaly using Cumulative Distribution Function (CDF), it is bounded using an error function into the interval from 0 to 1. The error function represents the approximate probability that the observation x_i drawn from random variable X lies in the range of $[-z_i, z_i]$ denoted as

$$E_A(z_i) = z_i \frac{e^{-z_i^2}}{\sqrt{\pi}} (2/1 + 4/3z_i^2 + 8/15z_i^4 + \dots). \quad (9)$$

CDF represents the probability that the random variable X takes a value less than or equal to x_i . $F_X: \mathbb{R} \rightarrow [0, 1]$. For generic normal distribution with sample mean $\bar{x}_n$ and sample deviation s_n the cumulative distribution function $F_X(x)$ equals to

$$F_X(x_i)_n = \frac{1}{2} (1 + E_A(\frac{z_i}{\sqrt{2}})). \quad (10)$$

Given the probability, we can also derive the value of x to which it belongs using a percent point function (PPF) for numerical approximation of inverse CDF (ICDF) denoted also as $F_X(x_i)_n^{-1}$.

PPF returns the threshold value for random variable X under which it takes a value less than or equal to the value, for which $F_X(x)$ takes probability lower than selected quantile q . $F_X^{-1}: [0, 1] \rightarrow \mathbb{R}$. An algorithm that calculates the value of the PPF is reported below as Algorithm 1.

Algorithm 1 Percent-Point Function for Normal Distribution

Input: quantile q , sample mean $\bar{x}_n$ (2), sample variance s_n^2 (4)

Output: threshold value $x_{n,q}$

Initialisation :

1: $f \leftarrow 10; l \leftarrow -f; r \leftarrow f;$

LOOP Process

2: **while** $F_X(l) - q > 0$ **do**

3: $r \leftarrow l;$

4: $l \leftarrow lf;$

5: **end while**

6: **while** $F_X(r) - q < 0$ **do**

7: $l \leftarrow r;$

8: $r \leftarrow rf;$

9: **end while**

10: $\hat{x}_{n,q} = \arg \min_z \|F_X(z) - q\| \text{ s.t. } l \leq z \leq r$

11: **return** $\hat{x}_{n,q} \sqrt{s_n^2} + \bar{x}_n$

D. Gaussian Anomaly Detection

Anomalies come in various kinds and flavors. Commonly denoted types are point (spatial), contextual, and collective (temporal) anomalies [2]. Spatial anomalies take on a value that particularly deviates from the sample mean $\bar{x}_n$. From a statistical viewpoint, spatial anomalies can be considered values x that significantly differ from the data distribution.

In empirical fields, such as machine learning, the three-sigma rule defines a region of distribution where normal values are expected to occur with near certainty. This assumption makes approximately 0.27% of values in the given distribution considered anomalous.

Definition 2.3 (Three-Sigma Rule of Thumb (3 σ rule)): 3σ rule represents a probability, that any value x_i of random variable X will lie within a region of values of normal distribution at the distance from the sample mean μ_n of at most 3 sample standard deviations σ_n .

$$P\{|x_i - \mu_n| < 3\sigma_n\} = 0.99730 \quad (11)$$

Anomalous values occur on both tails of the distribution. In order to discriminate the anomalies using the three-sigma rule on both tails of the distribution, we define the anomaly score as follows

$$y_i = 2 \left| F_X(x_i)_n - \frac{1}{2} \right|, \quad (12)$$

where

$$y_i \in [0, P\{|x_i - \mu_n| < 3\sigma_n\}], \quad (13a)$$

applies for normal observations and

$$y_i \in [P\{|x_i - \mu_n| < 3\sigma_n\}, 1], \quad (13b)$$

for anomalies.

Using pure statistics to model normal behavior lets us ask the question about the threshold value x which corresponds to the area under the curve of CDF equal to the given probability. A such query can be answered using inversion of (12). However, inversion of (12) would fail the horizontal line test. Therefore, we restrict the applicability of the inverse only to $F_X(x)_i \in [0.5, 1]$ and define upper threshold as follows

$$x_i = F_X \left(\frac{y_i}{2} + \frac{1}{2} \right)_n^{-1}. \quad (14)$$

In order to derive a lower threshold, the Gaussian distribution is fitted to the negative value of the streamed data and evaluated accordingly using the previously defined equations.

III. CONCLUSION

This paper proposes a novel approach to real-time anomaly detection that provides a physical threshold that bounds normal process operation. Such an approach has wide applicability in all the process automation fields where low latency evaluation and online adaptation are crucial. Moreover, adaptive operation

constraints provide less conservative process limits and govern important insight into systems behavior. The plug-and-play feature of the model makes it easily deployable as shown in two case studies.

The first case study performed on BESS examined the average battery cell temperature and demonstrated the ability to capture anomalies as well as the capacity to restrict the operational area by inversion of the cumulative distribution function. Following our investigation of state-of-the-art online anomaly detection described in Section I, we conclude, that although the robustness and performance of complex methods may exceed the performance of the proposed method, the ability to invert the prediction to depict real-time operational restrictions and eschew using non-comprehensible parameters makes it superior for a wide range of use cases. However, the performance might be greatly afflicted when the time constraints of the observed system are not known. This restriction is much weaker than the restriction of the need for data scientists skilled in the hyper-parameter tuning of unsupervised models. Moreover, hyper-parameter tuning calls for ground truth information about anomalies, which requires an exhaustive collection and is not possible in real-time.

Future work on this method will focus on multivariate online detection, varying positive and negative process limits, and automated system identification. These challenges aim to improve the method's performance and simplify its usage.

REFERENCES

- [1] J. Manyika, M. Chui, J. Bughin, R. Dobbs, P. Bisson, and A. Marrs, *Disruptive technologies: Advances that will transform life, business, and the global economy*. McKinsey Global Institute, 2013.
- [2] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, jul 2009. [Online]. Available: <https://doi.org/10.1145/1541880.1541882>
- [3] N. Laptev, S. Amizadeh, and I. Flint, "Generic and scalable framework for automated time-series anomaly detection," in *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '15. New York, NY, USA: Association for Computing Machinery, 2015, pp. 1939–1947. [Online]. Available: <https://doi.org/10.1145/2783258.2788611>
- [4] A. Kejariwal, "Introducing practical and robust anomaly detection in a time series," 2015. [Online]. Available: https://blog.twitter.com/engineering/en_us/a/2015/introducing-practical-and-robust-anomaly-detection-in-a-time-series
- [5] S. Ahmad and S. Purdy, "Real-time anomaly detection for streaming analytics," 2016. [Online]. Available: <https://arxiv.org/abs/1607.02480>
- [6] H. S. Pannu, J. Liu, and S. Fu, "Aad: Adaptive anomaly detection system for cloud computing infrastructures," in *2012 IEEE 31st Symposium on Reliable Distributed Systems*, 2012, pp. 396–397.
- [7] L. Zhang, J. Lin, and R. Karim, "Adaptive kernel density-based anomaly detection for nonlinear systems," *Knowledge-Based Systems*, vol. 139, pp. 50–63, 2018. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0950705117304707>
- [8] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and H. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys (CSUR)*, vol. 46, 04 2014.
- [9] H. H. Bosman, G. Iacca, A. Tejada, H. J. Wörtche, and A. Liotta, "Ensembles of incremental learners to detect anomalies in ad hoc sensor networks," *Ad Hoc Networks*, vol. 35, pp. 14–36, 2015, special Issue on Big Data Inspired Data Sensing, Processing and Networking Technologies. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1570870515001481>
- [10] S. Ahmad, A. Lavin, S. Purdy, and Z. Agha, "Unsupervised real-time anomaly detection for streaming data," *Neurocomputing*, vol. 262, pp. 134–147, 2017, online Real-Time Learning Strategies for Data Streams. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0925231217309864>
- [11] J. Liu, C. Su, X. Wang, W. Fang, S. Niu, and L. Cheng, "Abnormality in power system transient stability control of bess/statcom," *The Journal of Engineering*, vol. 2017, no. 13, pp. 1040–1044, 2017. [Online]. Available: <https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/joe.2017.0487>
- [12] X. Wang and S.-H. Ahn, "Real-time prediction and anomaly detection of electrical load in a residential community," *Applied Energy*, vol. 259, p. 114145, 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S030626191931832X>
- [13] Y. Dai, S. Sun, and L. Che, "Improved dbSCAN-based data anomaly detection approach for battery energy storage stations," in *Journal of Physics: Conference Series*, vol. 2351, 07 2022, p. 012025.
- [14] N. Schweikardt, *One-Pass Algorithm*. Boston, MA: Springer US, 2009, pp. 1948–1949. [Online]. Available: https://doi.org/10.1007/978-0-387-39940-9_253
- [15] B. P. Welford, "Note on a method for calculating corrected sums of squares and products," *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962. [Online]. Available: <https://www.tandfonline.com/doi/abs/10.1080/00401706.1962.10490022>